Fundamental
Of
AI and ML

Project Report

Name: Anshika Singh
Registration no.: 25BCE10770
Branch: B.tech CSE Core
Year: Semester 2 (1 st Year)
Course Code : CSA2001
Date: 30/03/2026

1. Introduction

In today's time, almost everyone uses mobile phones for communication due to the fact that the internet has become a necessity in our day to day life. Along with useful messages, people also receive unwanted messages like spam and scam messages. These messages often try to trick users by offering fake rewards, prizes, or urgent warnings like "your account is blocked" or "you have won a lottery." Many people, especially those who are not aware, may fall into these traps and lose money or personal information. So awareness plays an important role in it.

Because of this, it becomes important to have a system that can automatically detect such messages and warn users. This project focuses on building a **Scam Message Detector** using basic concepts of Artificial Intelligence and Machine Learning.

The system reads a message and predicts whether it is **safe (ham)** or **scam (spam)**. It uses simple Natural Language Processing (NLP) techniques to understand the text and a machine learning algorithm to classify it. This project shows how AI can be used in real life to improve digital safety.

2. Problem Statement

Nowadays, scam messages are increasing rapidly. It is not always easy for users to identify whether a message is genuine or fake. Some messages look very real and create urgency, which makes people trust them.

So, the problem is:

How can we automatically detect whether a message is spam or safe using machine learning?

Machine Learning helps solve this problem by analyzing message text and learning patterns from past data and helps to solve the problem step wise. By training a model on examples of spam and safe messages, the system can automatically predict whether a new message is a scam or not. This reduces human effort and helps in improving online safety.

3. Objective

The main goals of this project are:

- To build a simple model that can detect scam messages
- To learn and apply basic concepts of AI and ML
- To process text data using NLP techniques
- To classify messages into two categories: spam and ham
- To create a system where users can test their own messages
- To help people in differentiating spam messages and spread awareness regarding this.

4. Approach

The project is completed step by step in a simple way.

First, a dataset of messages is collected in a CSV file. Each message is labeled as either **spam** or **ham**. This data is used to train the model.

Next, the text data is cleaned. This includes:

- Converting all text to lowercase
- Removing numbers
- Removing punctuation (like commas, dots, etc.)

This step is important because it makes the data consistent and easier for the model to understand

After cleaning, the data is divided into two parts:

- **Training data** – used to train the model
- **Testing data** – used to check how well the model performs

Then, the text messages are converted into numbers using a method called **TF-IDF (Term Frequency-Inverse Document Frequency)**. Since machine learning models cannot understand text directly, this step helps in converting words into numerical values.

After that, a **Naive Bayes algorithm** is used to train the model. This algorithm is simple and works very well for text classification problems.

Finally, the model is tested, and accuracy is calculated. A user input feature is also added so that anyone can type a message and check whether it is scam or safe.

5. Tools & Technologies Used

The following tools , technologies and libraries are used in completion of the project:

- **Python** – main programming language
- **Pandas** – to load and manage data
- **Scikit-learn** – to build and tra
- **Re (Regular Expressions)** – for cleaning text
- **String module** – for removing punctuation
- **TF-IDF Vectorizer** – to convert text into numbers
- **Naive Bayes Algorithm** – for classification

6. Implementation

The implementation of this project is done in Python step by step.

First, the dataset is loaded using Pandas. Then the labels (spam and ham) are converted into numerical values (0 and 1).

After that, a function is created to clean the messages. This function removes unwanted characters and prepares the text for processing.

Next, the dataset is split into training and testing parts using `train_test_split`. Then, the TF-IDF vectorizer is applied to convert the messages into numerical form.

The Naive Bayes model is trained using the training data. Once the model is trained, it is used to predict the results for the test data.

The accuracy of the model is calculated using the `accuracy_score` function.

Finally, a loop is created where the user can enter any message, and the system will predict whether it is a scam or not.

7. Results

7.1 OVERVIEW :

The model works quite well and is able to identify most spam messages correctly. The accuracy depends on the dataset, but it generally gives good results. It will help user during any spam messages to identify whether it is safe or spam.

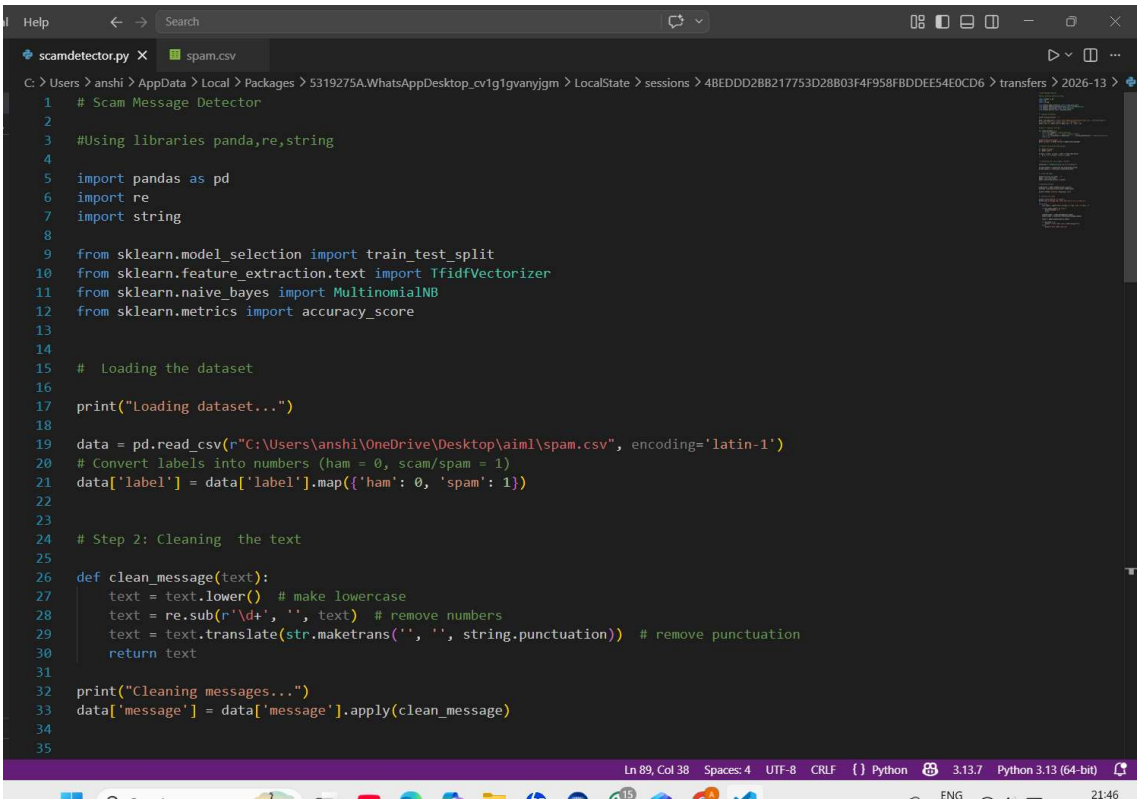
Some example outputs:

- Input: “You have won a free prize, click now”
  Output: Scam
- Input: “Let’s meet tomorrow for class”
  Output: Safe

This shows that the model can understand patterns like “win,” “free,” “offer,” and “urgent,” which are common in scam messages and detect it in a few seconds.

7.2 SCREENSHOTS ATTACHED :

INPUT CODE :



```python
# Scam Message Detector

#Using libraries panda,re,string

import pandas as pd
import re
import string

from sklearn.model_selection import train_test_split
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.naive_bayes import MultinomialNB
from sklearn.metrics import accuracy_score


#  Loading the dataset

print("Loading dataset...")

data = pd.read_csv(r"C:\Users\anshi\OneDrive\Desktop\aiml\spam.csv", encoding='latin-1')
# Convert labels into numbers (ham = 0, scam/spam = 1)
data['label'] = data['label'].map({'ham': 0, 'spam': 1})


# Step 2: Cleaning  the text

def clean_message(text):
    text = text.lower()  # make lowercase
    text = re.sub(r'\d+', '', text)  # remove numbers
    text = text.translate(str.maketrans('', '', string.punctuation))  # remove punctuation
    return text

print("Cleaning messages...")
data['message'] = data['message'].apply(clean_message)
```

scamdetector.py × spam.csv

C: > Users > anshi > AppData > Local > Packages > 5319275A.WhatsAppDesktop_cv1g1gvanyjgm > LocalState > sessions > 4BEDDD2BB217753D28B03F4F958FBDDEE54E0CD6 >

```python
data['message'] = data['message'].apply(clean_message)


# Prepare training and testing data

X = data['message']
y = data['label']

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)


 # Converting text into numbers (TF-IDF)

vectorizer = TfidfVectorizer(stop_words='english')

X_train_vectors = vectorizer.fit_transform(X_train)
X_test_vectors = vectorizer.transform(X_test)


#  Train the model

print("Training the model...")
model = MultinomialNB()
model.fit(X_train_vectors, y_train)


# Checking accuracy

predictions = model.predict(X_test_vectors)
accuracy = accuracy_score(y_test, predictions)

print(f"\nModel Accuracy: {accuracy:.2f}")
```

Ln 89, Col 38 Spaces: 4 UTF-8 CRLF {} Python 3.13.7

scamdetector.py spam.csv

C: > Users > anshi > AppData > Local > Packages > 5319275A.WhatsAppDesktop_cv1g1gvanyjgm > LocalState > sessions > 4BEDDD2BB217753D28

```python

predictions = model.predict(X_test_vectors)
accuracy = accuracy_score(y_test, predictions)

print(f"\nModel Accuracy: {accuracy:.2f}")


#  Testing with input

print("\nScam Detector is ready!")
print("Type a message and I will tell you if it's a scam.\n")

while True:
    user_input = input("Enter message (or type 'exit' to quit): ")

    if user_input.lower() == "exit":
        print("Goodbye 👋")
        break

    cleaned_input = clean_message(user_input)
    vector_input = vectorizer.transform([cleaned_input])

    result = model.predict(vector_input)

    if result[0] == 1:
        print("⚠️ This looks like a SCAM message!\n")
    else:
        print("✅ This seems safe.\n")
```

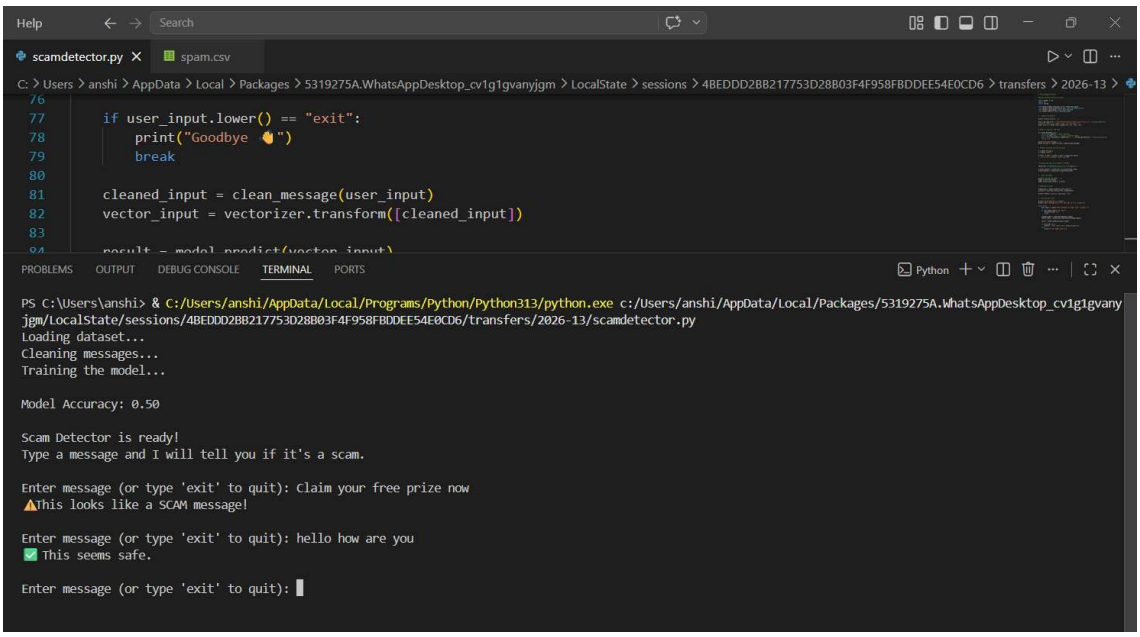
CSV FILE : DATASET OF MESSAGES

scamdetector.py spam.csv

C: > Users > anshi > OneDrive > Desktop > aiml > spam.csv > data

```csv
label,message
ham,Hello how are you
spam,Win money now
ham,Are you coming today
spam,Claim your free prize now
ham,Let's meet tomorrow
spam,You have won a lottery click here
```

OUTPUT CODE :


Help
Search
scamdetector.py
spam.csv
C: > Users > anshi > AppData > Local > Packages > 5319275A.WhatsAppDesktop_cv1g1gvanyjgm > LocalState > sessions > 4BEDDD2BB217753D28B03F4F958FBDDEE54E0CD6 > transfers > 2026-13 >
if user_input.lower() == "exit":
print("Goodbye 👋")
break
cleaned_input = clean_message(user_input)
vector_input = vectorizer.transform([cleaned_input])
PROBLEMS
OUTPUT
DEBUG CONSOLE
TERMINAL
PORTS
Python
PS C:\Users\anshi> & C:/Users/anshi/AppData/Local/Programs/Python/Python313/python.exe c:/Users/anshi/AppData/Local/Packages/5319275A.WhatsAppDesktop_cv1g1gvanyjgm/LocalState/sessions/4BEDDD2BB217753D28B03F4F958FBDDEE54E0CD6/transfers/2026-13/scamdetector.py
Loading dataset...
Cleaning messages...
Training the model...
Model Accuracy: 0.50
Scam Detector is ready!
Type a message and I will tell you if it's a scam.
Enter message (or type 'exit' to quit): Claim your free prize now
⚠️This looks like a SCAM message!
Enter message (or type 'exit' to quit): hello how are you
✅ This seems safe.
Enter message (or type 'exit' to quit):

8. Challenges Faced

While making this project, I faced some difficulties which helped me learn better.

One problem was that the **dataset was small**. Because of less data, the model was not always able to give correct results. I understood that more data helps the model learn better.

Another challenge was **cleaning the text data**. I had to remove numbers, punctuation, and make all text lowercase. But I had to be careful because removing too much can change the meaning of the message.

Some messages were also **confusing**. For example, messages like “special offer for you” can be real or fake. The model sometimes gets confused in such cases.

At first, I also found it difficult to understand how **text is converted into numbers using TF-IDF**, but after practicing, I understood how it works.

Overall, these challenges helped me improve my understanding of machine learning.

9. Future Improvements

This project can be improved in many ways in the future.

First, I can use a **bigger dataset** with more real-life messages. This will improve the accuracy of the model.

Second, I can add a **simple user interface (GUI)** so that users can easily use the system without typing commands.

I can also try **better algorithms** like Logistic Regression or deep learning models to get more accurate results.

Another improvement is to convert this into a **mobile app or website**, so people can use it anytime.

I can also add support for **different languages like Hindi**, so more people can use it.

10. Conclusion

In this project, I built a system that can detect whether a message is a scam or safe using machine learning.

I learned how to clean text data, convert it into numbers, and train a model to make predictions. I also understood how simple algorithms like Naive Bayes can solve real-life problems.

This project shows that even basic knowledge of AI and ML can be used to create useful applications.

Overall, this was a great learning experience for me, and I understood how AI can help in making our digital life safer.

11. References

- Scikit-learn Documentation: https://scikit-learn.org/stable/
- Pandas Documentation: https://pandas.pydata.org/
- Kaggle Datasets (Spam Dataset): https://www.kaggle.com/datasets
- Machine Learning Basics: https://www.geeksforgeeks.org/machine-learning/
- Naive Bayes Algorithm: https://www.geeksforgeeks.org/naive-bayes-classifiers/